

TA212: Manufacturing Processes – II

1st Semester 2025–26

Project Report on AIR DEFENCE SYSTEM

Group No: 5

Group Members

- | | |
|-----------------------|-----------------|
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Guide: Mr. Kishan Prajapati

Tutor: Mr. Amit Choudhary

Lab I/c.: Aman Singh

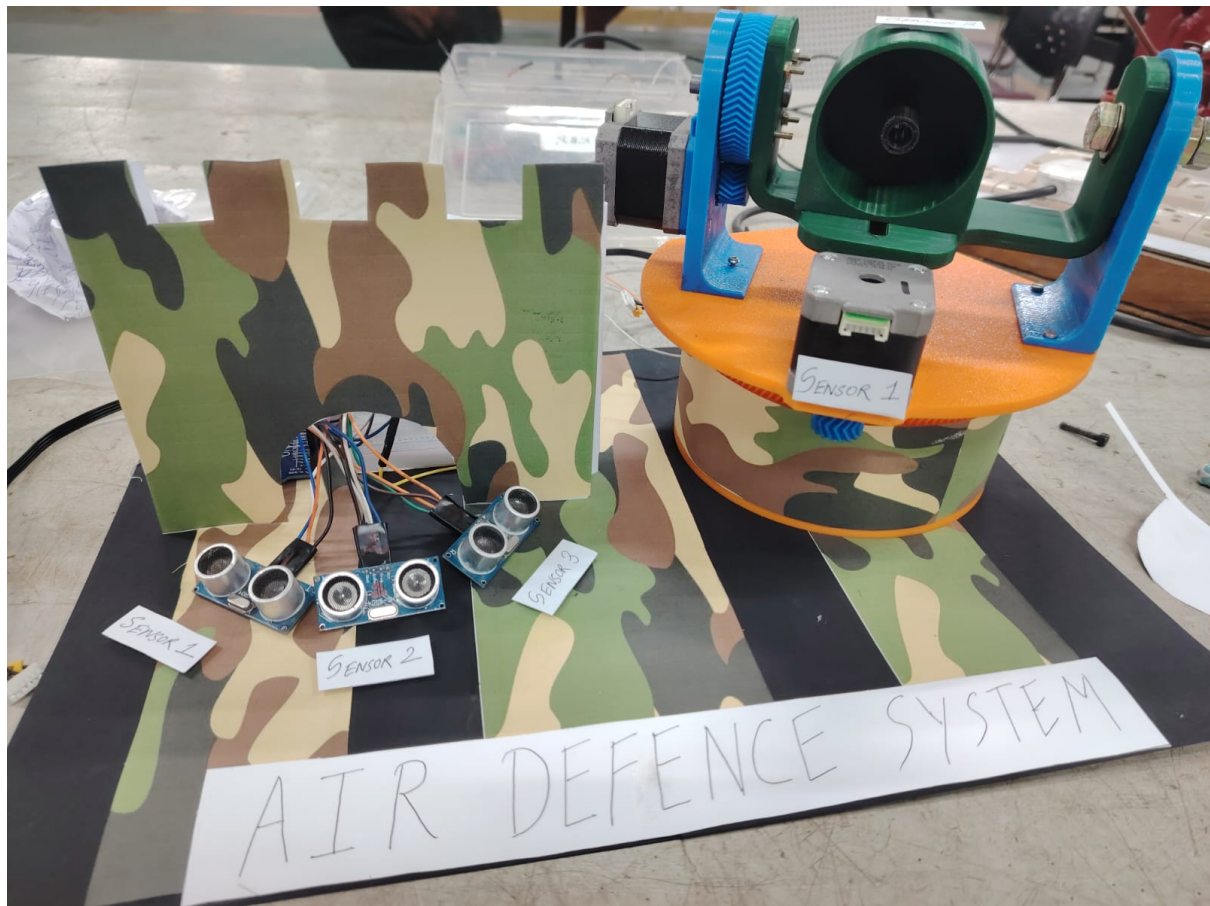
Course Instructor: Dr. Sarvesh Mishra

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Guide: Mr. Kishan Prajapati

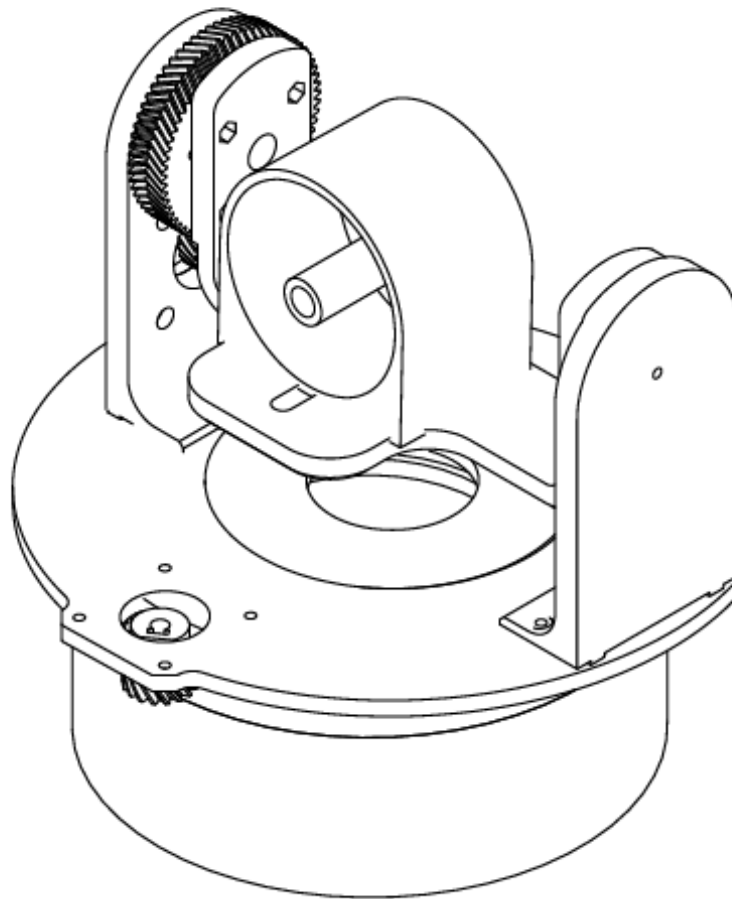
Tutor: Mr. Amit Choudhary

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Indian Institute of Technology, Kanpur

Isometric Drawing



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		Document type	Document status		
		Title Final1	DWG No.		
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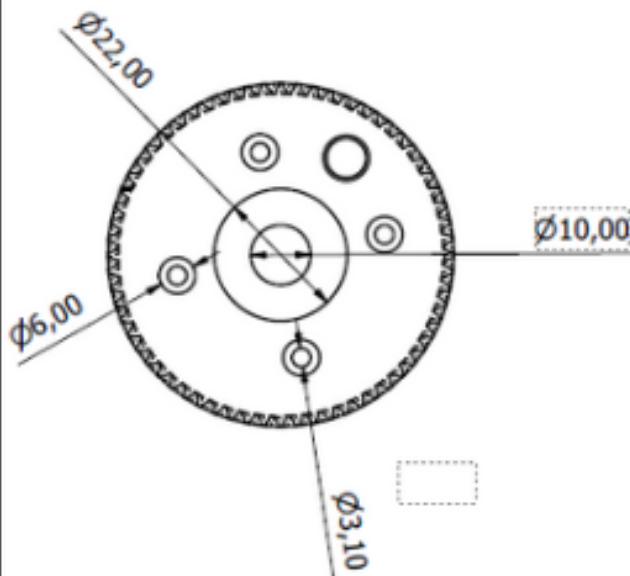
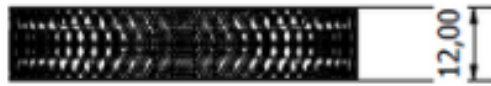
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Individual Parts' Information

Part Name	Material Used	Quantity
64 Teeth Herringbone	3D Printed	1
21 Teeth Herringbone	3D Printed	1
Bolt Spacer	3D Printed	1
Base Mount Bearing Clip Ring	3D Printed	1
Hall Effect Stepper Mount	3D Printed	1
Pan Mount	3D Printed	1
Idle Side Support	3D Printed	1
144 Teeth Herringbone Mount	3D Printed	1
17 Teeth Herringbone Gear	3D Printed	1
Gear Side Support	3D Printed	1
Trigger	3D Printed	1
Trigger Aligner	3D Printed	1
Pivot Support/Supporting Workpiece	Mild Steel	1

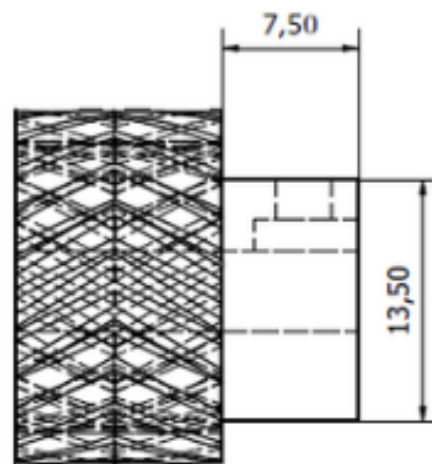
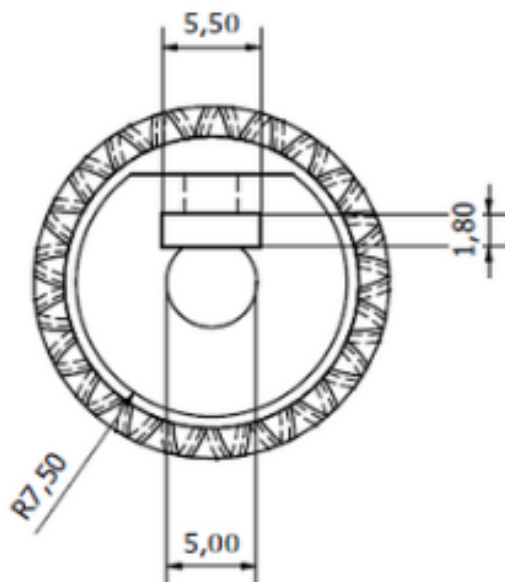
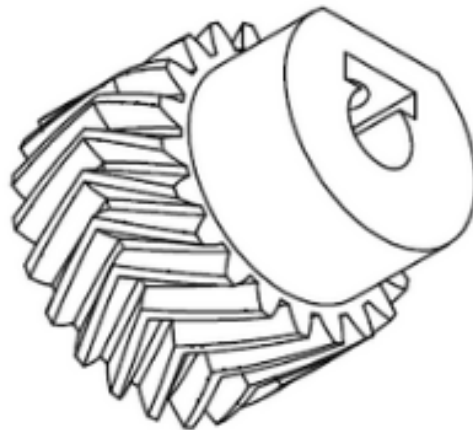
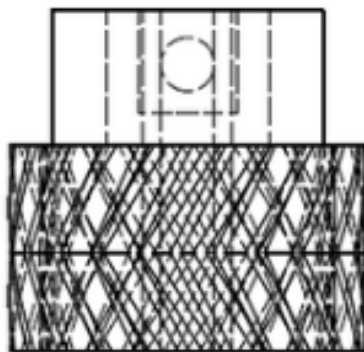
All Components' Part Drawings

64 TOOTH HERRINGBONE GEAR



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			SAMAYRAJ		
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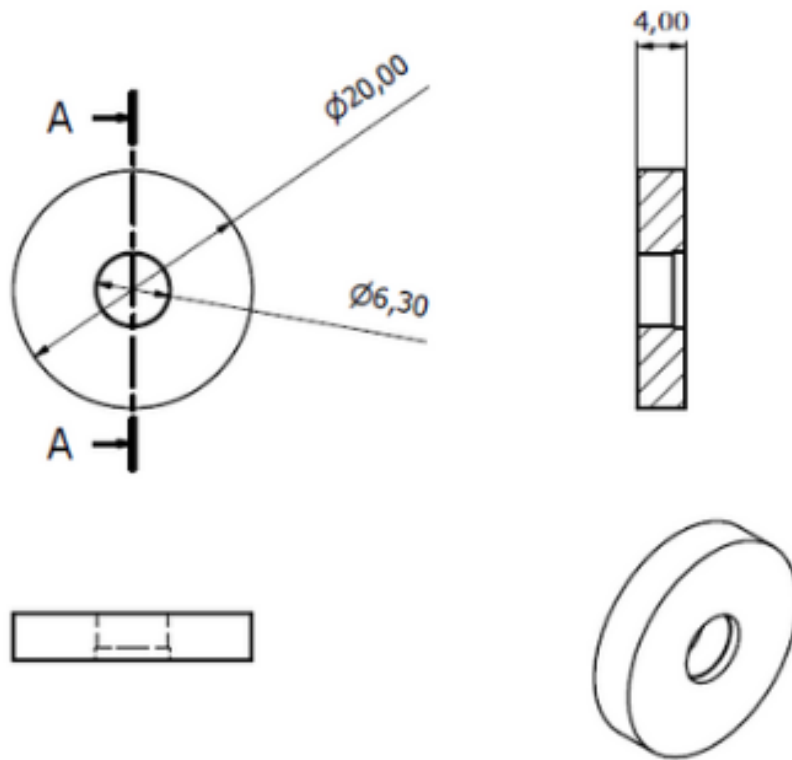
21 TOOTH HERRINGBONE GEAR



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			ASHWIN		
			21 Tooth Herringbone Gear	Edition	Sheet 1 / 1

A-A (2:1)

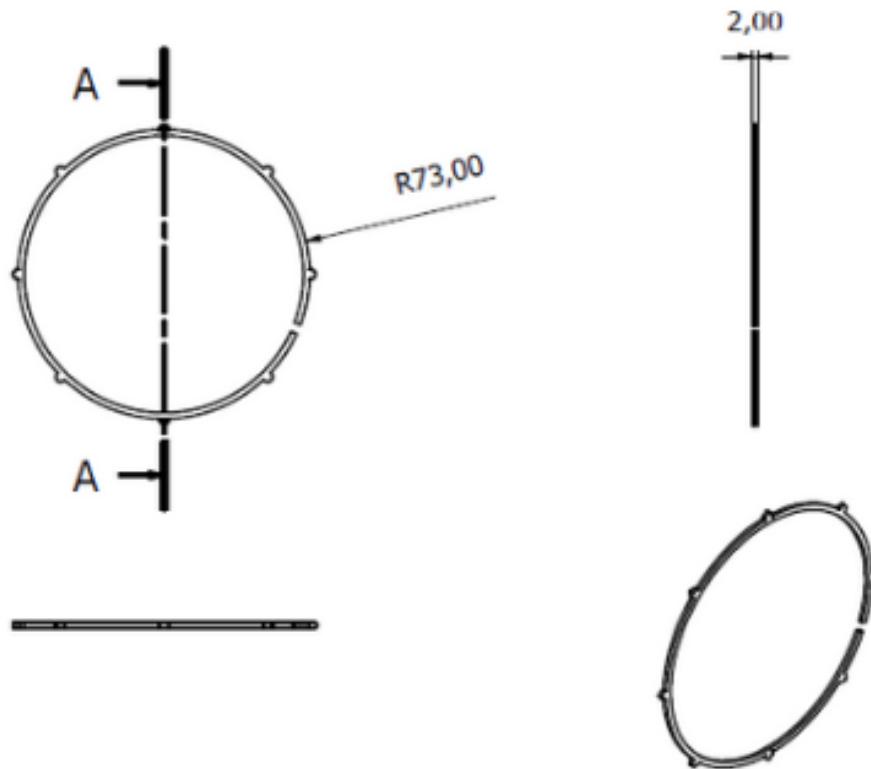
Bolt Spacer



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			TRISHA BAFNA		
			Camera Mounting Bolt Spacer	Edition	Sheet 1 / 1

A-A (1:3)

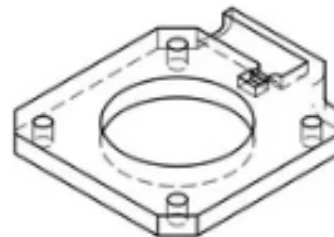
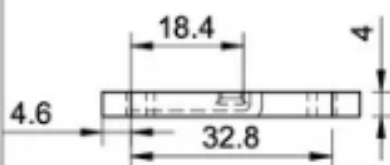
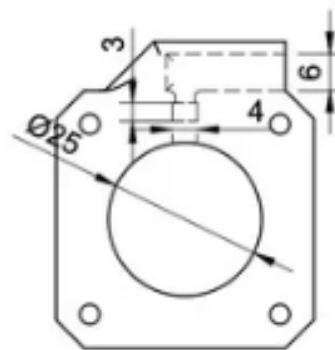
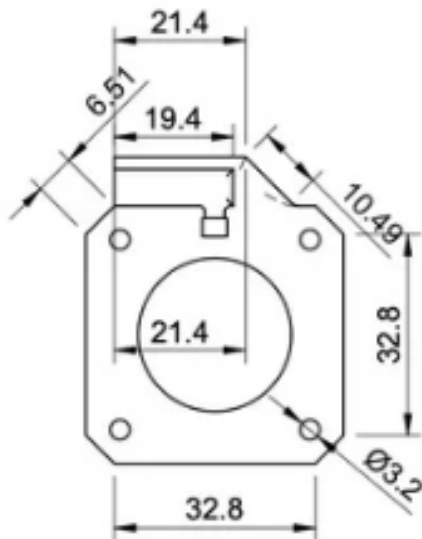
Base Mount Bearing Clip Ring



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			ABHINAV		
Base Mount Bearing Clip Ring				Edition	Sheet 1 / 1

HALL EFFECT STEPPER MOUNT

A-A 1:3



All dimensions are
in mm

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		Document type	Document status
		Title Hall Effect Stepper Mount	DWG No.
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Pan Mount

Top view

- Outer diameter: $\phi 130.80$
- Inner diameter: $\phi 126.80$
- Central hole diameter: $\phi 3.20$
- Top hole diameter: $\phi 23.00$
- Outer radius: $R98.37$
- Inner radius: $R77.00$
- Bottom radius: $R62.20$
- Side hole diameter: $\phi 3.00$
- Side hole radius: $R6.00$
- Side hole offset: $R5.00$
- Side hole offset: $R9.00$
- Side hole offset: 6.35
- Side hole offset: 5.00

Side view

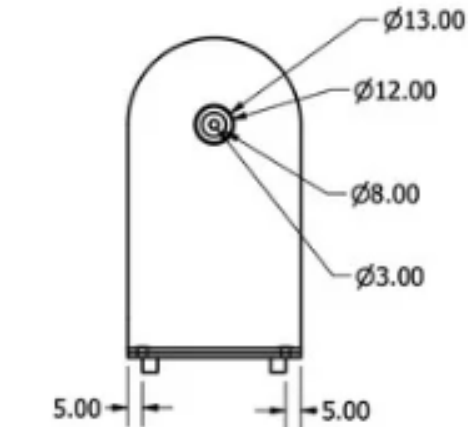
- Top flange thickness: 13.80
- Top flange offset: 8.80
- Bottom flange thickness: 5.00
- Bottom flange offset: 18.80

Isometric view

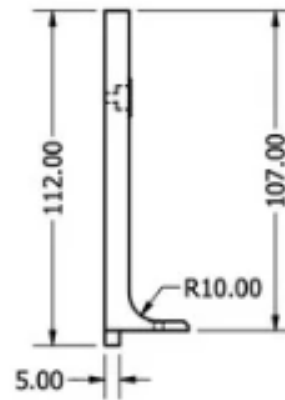
Scale - 1:2
All dimensions are in mm

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Pan Mount			Edition	Sheet 1 / 1

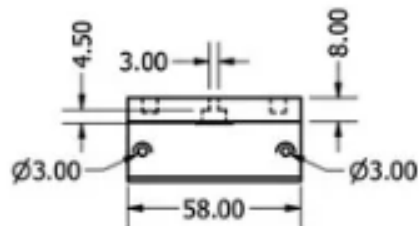
Idle Side Support



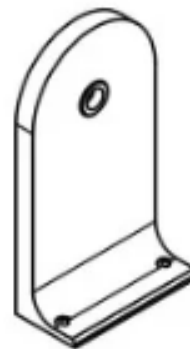
Front view



Side view



Top view

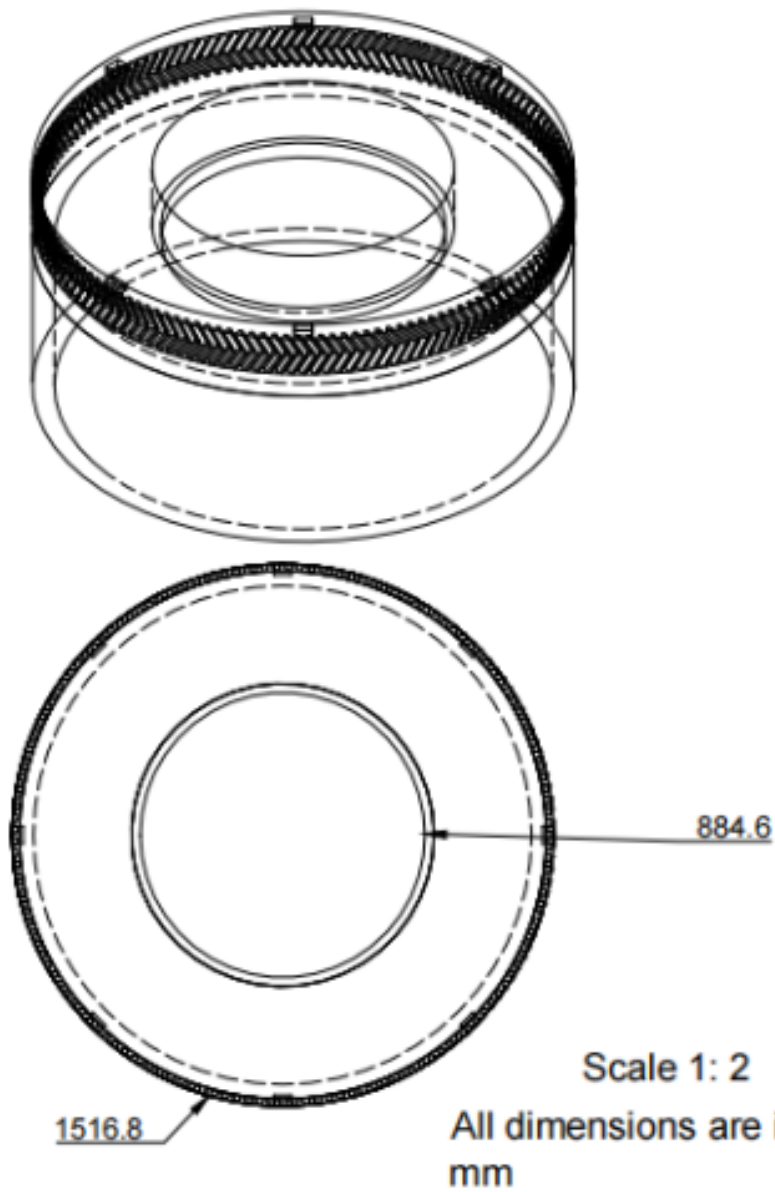


Isometric view

Scale - 1:2
All dimensions are in mm

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			Boddupally Utham Teja Sree 240280		
Idle Side Support				Edition	Sheet 1 / 1

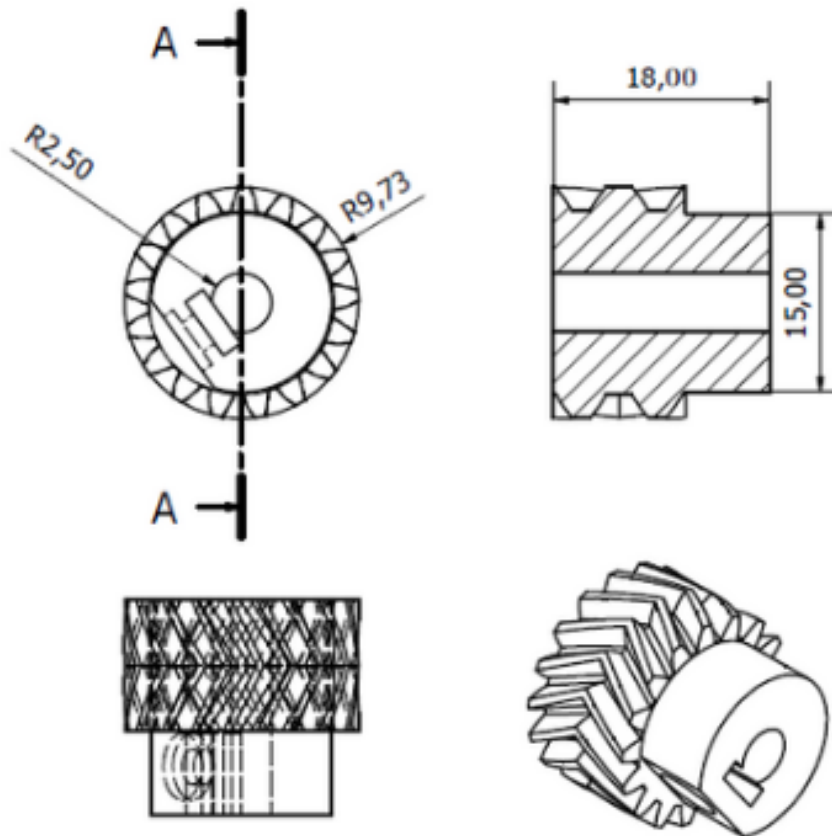
144 TOOTH HERRINGBONE MOUNT



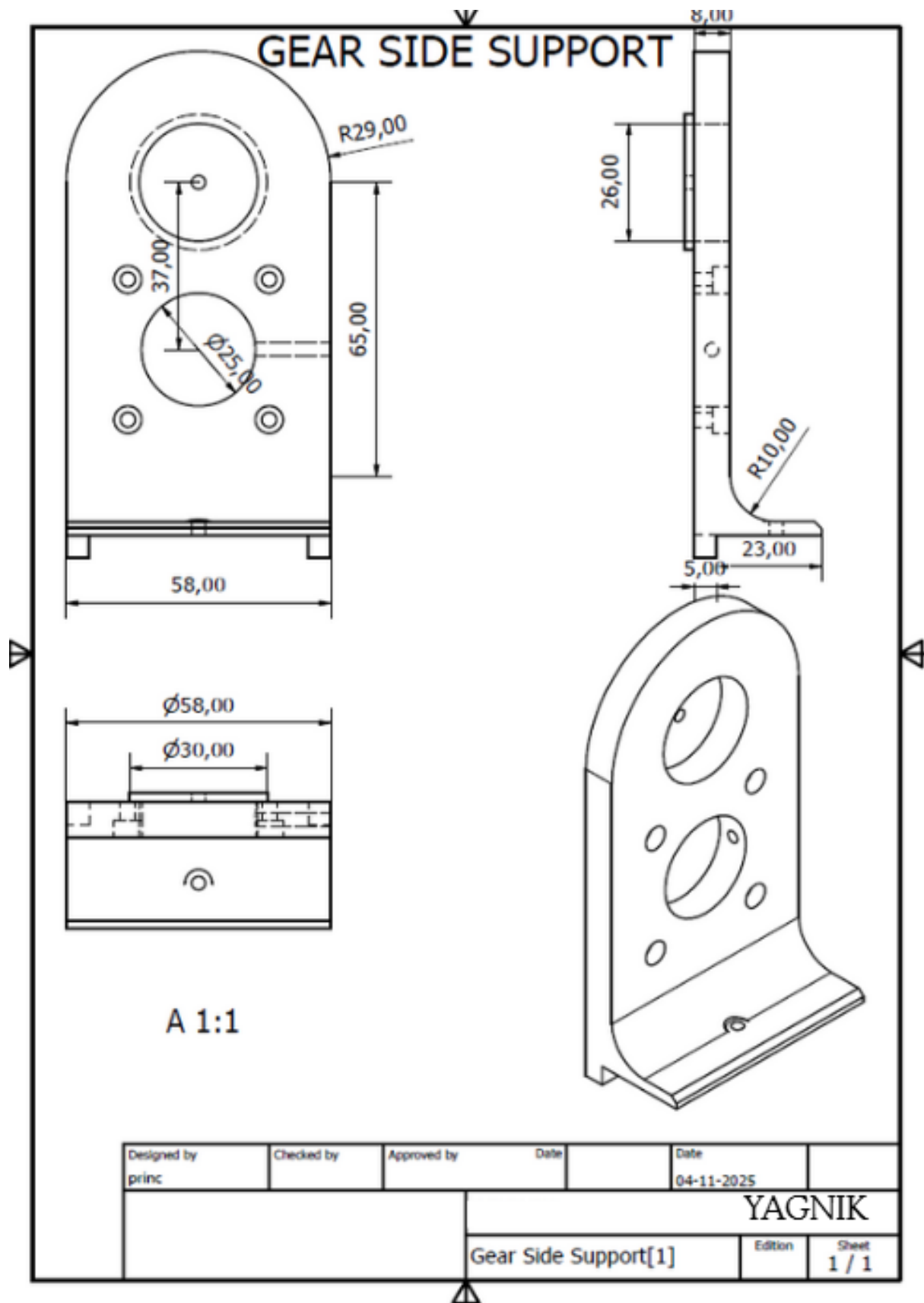
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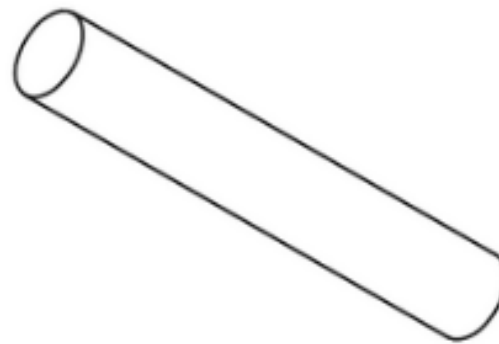
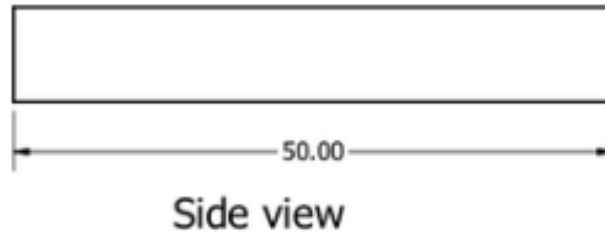
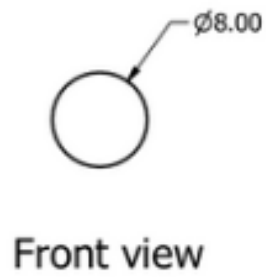
Tooth Herringbone Gear



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			TRISHA SHEKHAWAT		
			17 Tooth Herringbone Gear	Edition	Sheet 1 / 1



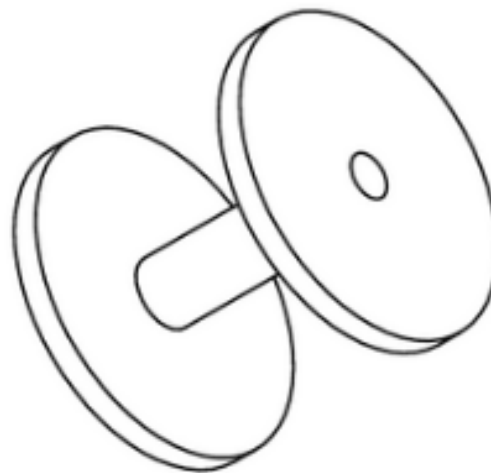
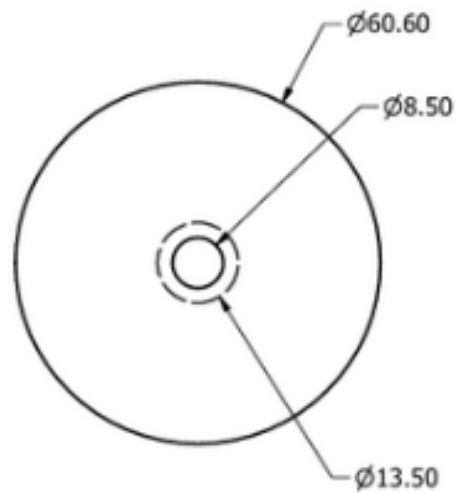
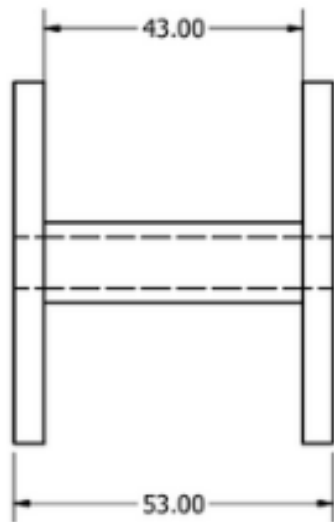
Trigger



All dimensions are in mm
Scale - 1:2

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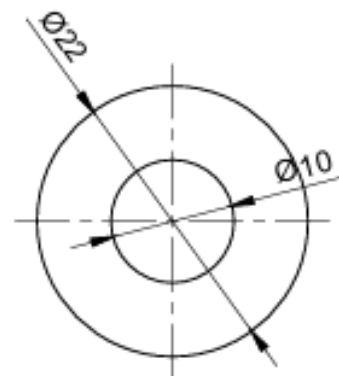
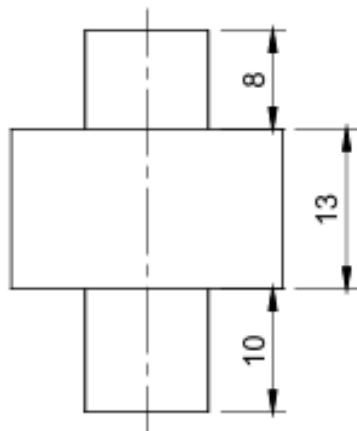
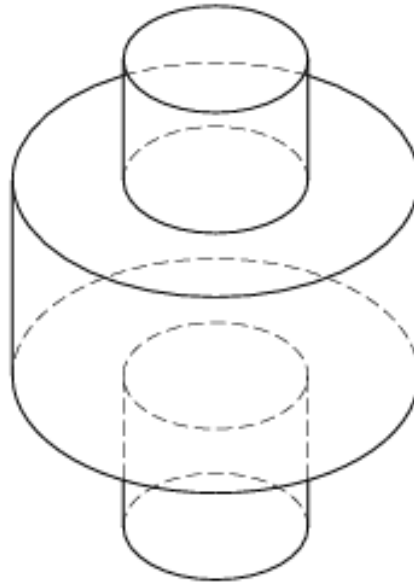
Trigger Aligner



All dimensions are in mm
Scale - 1:1

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SUPPORTING WORKPIECE



All dimensions in mm
Scale 3:1

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		Title WorkPiece	DWG No.		
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Project Overview

1. Introduction

The project aims to design and develop an automated air defense mechanism capable of detecting, tracking, and neutralizing a target within a specified range. The system integrates ultrasonic sensors, stepper motors, and a mechanical firing actuator, all coordinated through an Arduino microcontroller. This prototype demonstrates how sensor-based detection and automated targeting systems can be used in defense and surveillance applications.

2. Objectives

- a) To detect an approaching object using ultrasonic sensors.
- b) To calculate the relative position (distance/coordinates) of the detected object.
- c) To orient the turret automatically toward the target using two-axis stepper motor control.
- d) To launch a projectile using an mechanical firing actuator once the target is aligned.
- e) To demonstrate a low-cost automated defense system for small-scale demonstration or educational use.

2. System Components

1. Ultrasonic Sensors (3 units):
 - One placed on the system.
 - One on tilt mechanism.
 - One on pan mechanism.
 - Used to detect the distance of the target and estimate its position in 3D space.

2. Arudino Microcontroller (Arduino UNO):

- Acts as the brain of the system.
- Processes sensor data and controls the rotation and firing mechanisms.

3. Stepper Motors (2 units):

- One controls horizontal rotation (Y-axis).
- One controls vertical tilt (Z-axis).
- Provides precise positioning of the turret.

4. Power Supply:

- Provides sufficient current to the motors.
- Usually consists of separate regulated supplies for control and power circuits.

5. Driver Modules:

- Motor Driver (DRV8255): Controls stepper motor movement.

4. Working Principle

1. **Detection:**

The ultrasonic sensors continuously emit and receive sound waves to measure distances and detect the object.

2. **Position Calculation:**

We have a system of 3 ultrasonic sensors continuously rotating. When an object is detected in any of them, the degree to which the motors have to move is calibrated by the code.

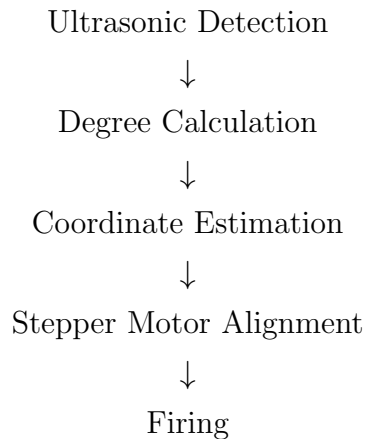
3. **Targeting:**

The Arduino sends control signals to the two stepper motors, rotating the turret to align with the detected coordinates.

4. **Firing Mechanism:**

Once aligned, the Arduino stops the turret in the target's direction. The firing is then done manually by the operator using a mechanical trigger to launch the projectile.

5. Control Flow



Acknowledgement

We would like to express our sincere gratitude to our guide, Mr. Kishan Prajapati sir, for his continuous support, valuable guidance and encouragement throughout this project. We also thank our department faculty and laboratory staff for providing the facilities and resources necessary to complete this work. Finally, we extend our appreciation to our teammates for their cooperation, ideas and teamwork in developing our Automated Air Defence Mechanism project.

Abstract

This project presents the design and development of an Automated Air Defence Mechanism capable of detecting, tracking and firing at a target automatically. The system uses three ultrasonic sensors placed on the system that keep rotating continuously and detect static objects. Based on the calculated degrees, two stepper motors rotate the turret along the Y and Z axes to align with the target. Once aligned, a mechanical firing mechanism launches an aluminium projectile toward the target. The prototype demonstrates the integration of sensor-based detection, automated motion control, and actuation for defense applications, providing a foundation for future advancements in autonomous targeting systems.

Motivation

Our motivation behind this project is to design a simple, low-cost prototype that demonstrates the basic working principle of an air defence system — using Arduino, sensors and motors to detect an approaching object and automatically aim or respond.

Through this project, we aimed to:

- Understand how sensors can detect motion or objects in real-time.
- Learn how Arduino can control motors to simulate target tracking.
- Explore the integration of hardware and programming to create an automated defence mechanism.
- Develop an interest in defence technology and robotics applications.

Ultimately, our goal was to create a miniaturized demonstration of an intelligent air defence system, showcasing how automation and embedded systems can play a vital role in national security.

Motor Sizing: Motor torque requirement calculation

Calculation based on provided data and general assumptions

I. Tilt Motor

1. Given Data and Assumptions

This calculation is based on the data provided and the following general assumptions for missing values.

- $m_{\text{load}} = 70.11 \text{ kg} = 0.07191 \text{ kg}$ (Load mass)
- $m_{\text{gear}} = 15 \text{ g} = 0.015 \text{ kg}$ (Driven gear mass)
- $R_1 = 7.5 \text{ mm} = 0.0075 \text{ m}$ (Driving gear radius)
- $R_2 = 11.5 \text{ mm} = 0.0115 \text{ m}$ (Driven gear radius)
- $d_{\text{sg}} = 2.25 \text{ cm} = 0.025 \text{ m}$ (Assumed C.G. distance)
- $\mu = 0.35$ (Assumed friction coeff. PLA-PLA)
- $R_{\text{axle}} = 5 \text{ mm} = 0.005 \text{ m}$ (Assumed axle radius)
- $V = 60 \text{ rpm}$ (Assumed operating speed)
- $t_g = 1.0$ (Assumed gear ratio)
- $F_{\text{OS}} = 2.0$ (Factor of Safety, from PDF)
- $g = 9.81 \text{ m/s}^2$ (Gravitational acceleration)

2. Step-by-Step Calculation

The total torque required from the motor is the sum of the load, friction, and acceleration torques, all reflected back to the motor via the gear ratio and multiplied by a Factor of Safety (FOS).

$$T_{\text{total}} = (T_L + T_F + T_A) \times \text{FOS}$$

2.1 Gear Ratio (i)

The gear ratio (i) reflects the torque from the load shaft back to the motor shaft.

$$i = \frac{R_2}{R_1} = \frac{0.0075 \text{ m}}{0.0115 \text{ m}} = 0.652$$

2.2 Load Torque (T_L) (due to Gravity)

First, calculate the force and torque at the load's arm.

$$F_L = m_{\text{load}} \times g = 0.07191 \text{ kg} \times 9.81 \text{ m/s}^2 = 0.776 \text{ N}$$

$$T_{L,\text{load}} = F_L \times d_{\text{sg}} = 0.776 \text{ N} \times 0.025 \text{ m} = 0.0194 \text{ Nm}$$

Next, reflect this torque back to the motor.

$$T_L = T_{L,\text{load}} \times i = 0.0194 \text{ Nm} \times 0.652 = 0.0126 \text{ Nm}$$

2.3 Friction Torque (T_f)

First, find the total normal force (F_N) from the combined weight of the load and the driven gear.

$$m_{\text{total}} = m_{\text{load}} + m_{\text{gear}} = 0.07191 \text{ kg} + 0.015 \text{ kg} = 0.09411 \text{ kg}$$

$$F_N = m_{\text{total}} \times g = 0.09411 \text{ kg} \times 9.81 \text{ m/s}^2 = 0.923 \text{ N}$$

Next, find the friction torque at the axle.

$$T_{f,\text{load}} = (F_N \times \mu) \times R_{\text{axle}} = (0.35 \times 0.923 \text{ N}) \times 0.005 \text{ m} = 0.0016 \text{ Nm}$$

Finally, reflect this torque back to the motor.

$$T_f = T_{f,\text{load}} \times i = 0.0016 \text{ Nm} \times 0.652 = \mathbf{0.0010 \text{ Nm}}$$

2.4 Acceleration Torque (T_a)

The acceleration torque is based on the total reflected inertia (J_{total}) and the formula from the lecture PDF.

$$J_{\text{total}} = 2.83 \times 10^{-5} \text{ kg}\cdot\text{m}^2 \quad (\text{Calculated inertia reflected to motor})$$

$$T_a = \frac{J_{\text{total}} \times V}{9.55 \times t_a} = \frac{(2.83 \times 10^{-5} \text{ kg}\cdot\text{m}^2) \times (60 \text{ rpm})}{9.55 \times 1.0 \text{ s}}$$

$$T_a = \mathbf{0.00018 \text{ Nm}} \quad (\text{This value is negligible})$$

3. Total Required Torque (T_{total})

Sum the components and apply the Factor of Safety.

$$T_{\text{total}} = (T_L + T_f + T_a) \times \text{FOS}$$

$$T_{\text{total}} = (0.0126 + 0.0010 + 0.00018) \text{ Nm} \times 2.0$$

$$T_{\text{total}} = (0.01378 \text{ Nm}) \times 2.0$$

$$\boxed{T_{\text{total}} = 0.0276 \text{ Nm}}$$

Conclusion

The total estimated torque required from the motor is **0.0276 Nm** (or 2.76 N·cm). This is well within the typical holding torque capacity of a standard NEMA 17 motor.

II. Pan Motor

1. Given Data and Assumptions

This calculation is for the bottom motor rotating the entire upper assembly (modeled as an “Index Table”).

- $m = 634 \text{ g} = 0.634 \text{ kg}$ (Total rotating mass)
- $R = 63 \text{ mm} = 0.063 \text{ m}$ (Radius of assembly, as a disc)
- $\mu = 0.3$ (Assumed bearing friction coeff.)

- $R_{\text{axle}} = 5 \text{ mm} = 0.005 \text{ m}$ (Assumed bearing axle radius)
- $V = 60 \text{ rpm}$ (Assumed operating speed)
- $t_a = 1.0 \text{ s}$ (Assumed acceleration time)
- $\text{FOS} = 2.0$ (Factor of Safety, from PDF)
- $g = 9.81 \text{ m/s}^2$ (Gravitational acceleration)

2. Step-by-Step Calculation

For a horizontally rotating table, the total torque is the sum of the friction torque and the acceleration torque, multiplied by a Factor of Safety (FOS). There is no load torque due to gravity.

$$T_{\text{total}} = (T_f + T_a) \times \text{FOS}$$

2.1 Friction Torque (T_f)

This is the torque required to overcome the friction in the central rotating bearing. First, calculate the Normal Force (F_N) from the total weight of the assembly.

$$F_N = m \times g = 0.634 \text{ kg} \times 9.81 \text{ m/s}^2 = 6.22 \text{ N}$$

Next, find the friction torque at the bearing.

$$T_f = (\mu \times F_N) \times R_{\text{axle}} = (0.3 \times 6.22 \text{ N}) \times 0.005 \text{ m}$$

$$T_f = \mathbf{0.0093 \text{ Nm}}$$

2.2 Acceleration Torque (T_a)

This is the torque required to accelerate the assembly from rest to its operating speed. First, calculate the Moment of Inertia (J_d) for the assembly, modeling it as a solid disc (as per the PDF's "Index Table" example).

$$J_d = \frac{1}{2} m R^2 = 0.5 \times (0.634 \text{ kg}) \times (0.063 \text{ m})^2$$

$$J_d = 0.00126 \text{ kg}\cdot\text{m}^2$$

Next, use the formula from the lecture PDF to find the acceleration torque.

$$T_a = \frac{J_d \times V}{9.55 \times t_a} = \frac{(0.00126 \text{ kg}\cdot\text{m}^2) \times (60 \text{ rpm})}{9.55 \times 1.0 \text{ s}}$$

$$T_a = \mathbf{0.0079 \text{ Nm}}$$

3. Total Required Torque (T_{Total})

Sum the components and apply the Factor of Safety.

$$T_{\text{Total}} = (T_{\text{F}} + T_{\text{R}}) \times \text{FOS}$$

$$T_{\text{Total}} = (0.0035 \text{ Nm} + 0.0079 \text{ Nm}) \times 2.0$$

$$T_{\text{Total}} = (0.0172 \text{ Nm}) \times 2.0$$

$$\boxed{T_{\text{Total}} = 0.0344 \text{ Nm}}$$

Conclusion: The total estimated torque required from the bottom motor is 0.0344 Nm (or 3.44 mNm). This is well within the typical holding torque capacity of a standard NEMA 17 motor.

Cost Estimation

Costs Given:

Material Cost (/ Kg):

- Mild Steel : Rs. 90
- Nuts and Bolts : Rs. 130
- Clips and Fold : Rs. 120

Machine Usage Cost (/ Hrs.):

- 3D Printing : Rs. 300
- Lathe : Rs. 350
- Milling : Rs. 450
- Drilling : Rs. 100
- Cutting : Rs. 60

Labour Cost (/Day or /8 Hrs.):

- Unskilled : Rs. 650
- Skilled : Rs. 850

Cost of Electronics Kit:

- With single Motor : Rs. 2500
- With 2 Motors : Rs. 3000

Material Weight

Mild Steel: 1 piece

$$\begin{aligned}\text{Volume} &= \pi \times 10 \times 10 + \pi \times 5 \times 5 \times 10 + \pi \times 13 \times 5 \times 5 \\ &= \pi \times 1375 \\ &= 4319.68 \text{ mm}^3\end{aligned}$$

Density of Mild steel = 7850 kg/m^3

$$\begin{aligned}\text{Weight} &= \text{Volume} \times \text{Density} \\ &= 0.339 \text{ kg} = 339 \text{ gm}\end{aligned}$$

Nuts and Bolts: 7 M2 + 2 M10 + 2 Big Nuts + 1 Washer + 4 Nuts

$$7 \times 5 \text{ gm} + 2 \times 12 \text{ gm} + 2 \times 4 \text{ gm} + 1 \times 2 \text{ gm} + 4 \times 1.3 \text{ gm} = 74.2 \text{ gm}$$

Clips and Fold: 0 kg

Machine Usage Time

3D Printing: $3 \times 3 + 2.5 \times 2 + 2 \times 1 = 16$ hours

Lathe: 1 hour 15 min

Milling: 0 hours

Drilling: 30 mins

Cutting: 0 hours

Labour Hours

Unskilled: $13 \text{ Labs} + 2 \times 10 + 1 \times 8 = 13 \times 3 + 2 \times 10 + 1 \times 8 = \mathbf{67 \text{ Hours}}$

Electronics Kit

2 Motors used

Material Cost

Mild Steel: $Rs. 90 /Kg \times 0.339 \text{ kg} = Rs. 30.51$

Nuts and Bolts: $Rs. 130 /Kg \times 0.0742 \text{ kg} = Rs. 9.64$

Clips and Fold: $Rs. 0$

Machine Usage Time

3D Printing: $Rs. 300 /Hrs. \times 16 \text{ Hrs.} = Rs. 4800$

Lathe: $Rs. 350 /Hrs. \times 1.25 \text{ Hrs.} = Rs. 437.5$

Milling: $Rs. 0$

Drilling: $Rs. 100 /Hrs. \times 0.5 \text{ Hrs.} = Rs. 50$

Cutting: $Rs. 0$

Labour Hours

Unskilled: $Rs. 650 /8Hrs. \times 67 \text{ Hrs.} = Rs. 5443.75$

Electronics Kit

2 Motors used: $Rs. 3000$

Sensors: $Rs. 960$

Springs: $Rs. 320$

Driver: $Rs. 560$

Mosfets: $Rs. 240$

Total : Rs. 15,831.4

Thankyou - Jai Hind